

Daksh — Class X Study Test

Subject: Science | **Test:** Test 1 (Day 7, 16 May 2026) — Weekly
Time Allowed: 3 Hours • Maximum Marks: 80

Syllabus covered: Chemical Reactions and Equations • Acids, Bases and Salts • Metals and Non-metals

General Instructions:

- (i) This question paper contains five sections — A, B, C, D and E.
- (ii) Section A has Multiple Choice / Very Short Answer Questions of 1 mark each.
- (iii) Section B has Short Answer Questions of 2 marks each.
- (iv) Section C has Short Answer Questions of 3 marks each.
- (v) Section D has Long Answer Questions of 5 marks each.
- (vi) Section E has Case-based / Source-based questions of 4 marks each.
- (vii) All questions are compulsory. Internal choice has been provided in some questions wherever indicated.
- (viii) Use of calculators is not permitted.

SECTION A — Multiple Choice / Very Short Answer Questions (20 × 1 = 20 marks)

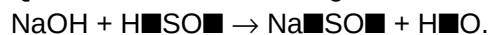
Choose the correct option or answer in one or two words.

- Q1.** The chemical equation $\text{Fe} + \text{CuSO}_4 \rightarrow \text{FeSO}_4 + \text{Cu}$ represents which type of reaction? (a) Combination (b) Decomposition (c) Displacement (d) Double displacement
- Q2.** Which of the following is a balanced chemical equation? (a) $\text{H}_2 + \text{O}_2 \rightarrow \text{H}_2\text{O}$ (b) $2\text{H}_2 + \text{O}_2 \rightarrow 2\text{H}_2\text{O}$ (c) $\text{H}_2 + 2\text{O}_2 \rightarrow 2\text{H}_2\text{O}$ (d) $2\text{H}_2 + 2\text{O}_2 \rightarrow \text{H}_2\text{O}$
- Q3.** Rancidity in oily food can be prevented by: (a) adding antioxidants (b) keeping food in transparent jars (c) increasing temperature (d) adding water
- Q4.** When silver chloride is exposed to sunlight, it turns: (a) yellow (b) brown (c) grey (d) black
- Q5.** Respiration is an example of: (a) endothermic reaction (b) exothermic reaction (c) physical change (d) decomposition only
- Q6.** The pH of a neutral solution is: (a) 0 (b) 1 (c) 7 (d) 14
- Q7.** The acid present in lemon juice is: (a) acetic acid (b) citric acid (c) lactic acid (d) tartaric acid
- Q8.** Bleaching powder is prepared by passing chlorine over: (a) NaOH (b) dry slaked lime (c) gypsum (d) baking soda
- Q9.** Which of the following salts does NOT contain water of crystallisation? (a) Blue vitriol (b) Baking soda (c) Washing soda (d) Gypsum
- Q10.** A solution turns red litmus blue. Its pH is likely to be: (a) 1 (b) 4 (c) 5 (d) 10
- Q11.** Which one of the following is the most reactive metal? (a) Copper (b) Iron (c) Sodium (d) Zinc
- Q12.** The metal that exists in liquid state at room temperature is: (a) mercury (b) sodium (c) potassium (d) gallium
- Q13.** Which of the following pairs will give displacement reactions? (a) $\text{NaCl} + \text{Cu}$ (b) $\text{MgCl}_2 + \text{Al}$ (c) $\text{FeSO}_4 + \text{Ag}$ (d) $\text{AgNO}_3 + \text{Cu}$
- Q14.** Which gas is liberated when an active metal reacts with a dilute acid? (a) Oxygen (b) Hydrogen (c) Chlorine (d) Carbon dioxide
- Q15.** The non-metal that conducts electricity is: (a) sulphur (b) phosphorus (c) graphite (d) iodine
- Q16.** Aqua regia is a mixture of: (a) $\text{HCl} + \text{HNO}_3$ (3:1) (b) $\text{HCl} + \text{HNO}_3$ (1:3) (c) $\text{HCl} + \text{H}_2\text{SO}_4$ (1:1) (d) $\text{HNO}_3 + \text{H}_2\text{SO}_4$ (3:1)
- Q17.** Define corrosion in one sentence.
- Q18.** Why is sodium kept immersed in kerosene oil?
- Q19.** Name the gas evolved when zinc granules are added to dilute hydrochloric acid.
- Q20.** Write the chemical formula of plaster of Paris.

SECTION B — Short Answer-I (6 × 2 = 12 marks)

Answer each question in 30–50 words.

Q21. Balance the following chemical equation and identify the type of reaction:



Q22. Why is respiration considered an exothermic reaction? Explain with a balanced equation.

Q23. What is meant by the pH of a solution? How does it affect dental health?

Q24. Why does dry hydrogen chloride gas not turn blue litmus red, whereas its aqueous solution does?

Q25. Define amphoteric oxide. Give one example with the relevant equations.

Q26. Why are ionic compounds usually solids with high melting and boiling points?

SECTION C — Short Answer-II (7 × 3 = 21 marks)

Answer each question in 50–80 words.

Q27. Distinguish between combination and decomposition reactions with one balanced equation each.

Q28. Explain the term 'redox reaction' with a balanced example. Identify the species oxidised and reduced.

Q29. Give the chemical name and one important use of: (i) baking soda, (ii) washing soda, (iii) bleaching powder.

Q30. What happens when (with balanced equations): (i) sodium hydroxide reacts with hydrochloric acid; (ii) zinc oxide reacts with dilute hydrochloric acid; (iii) carbon dioxide is passed through lime water?

Q31. Compare the physical properties of metals and non-metals (any three points).

Q32. Describe with an example how the reactivity series of metals helps in extracting metals.

Q33. Why do metals like copper not react with dilute hydrochloric acid, whereas iron does? Write the relevant equation for iron.

SECTION D — Long Answer (3 × 5 = 15 marks)

Answer each question in detail.

Q34. (a) Define the following terms with one example each: (i) oxidation, (ii) reduction, (iii) corrosion. (3)

(b) Explain why painting iron articles prevents rusting. (1)

(c) State one advantage and one disadvantage of galvanisation. (1)

Q35. (a) Define a neutralisation reaction with one balanced example. (1)

(b) Explain the role of HCl in our stomach. What happens when its level increases beyond normal? Suggest one antacid commonly used. (2)

(c) Common salt is the raw material for several chemicals. List any two such chemicals and write the chemical reaction for any one. (2)

Q36. (a) What is meant by the activity series of metals? Arrange Cu, Fe, Mg, Zn, Au in decreasing order of reactivity. (2)

(b) Explain why food cans are coated with tin and not zinc. (1)

(c) Write balanced chemical equations for the reactions of: (i) magnesium with steam, (ii) iron with chlorine. (2)

SECTION E — Case-based / Source-based (3 × 4 = 12 marks)

Read each case and answer the sub-parts.

Q37. Case Study (Chemical Reactions). A student takes a small amount of lead nitrate solution in a test tube and adds a few drops of potassium iodide solution to it. A bright yellow precipitate appears.

(i) Write the balanced chemical equation for the reaction. (1)

(ii) Identify the type of reaction. (1)

(iii) Name the yellow precipitate formed. (1)

(iv) Why does this reaction take place spontaneously? (1)

Q38. Case Study (Acids, Bases and Salts). Ramesh's grandmother applies a mixture of baking soda paste to her hands when she handles chillies, while she uses lemon juice to remove tea stains.

(i) Why does baking soda relieve the burning sensation? (1)

(ii) Why does lemon juice remove tea stains? (1)

(iii) Write the chemical name and formula of baking soda. (1)

(iv) Give one other household use of baking soda. (1)

Q39. Case Study (Metals and Non-metals). Anjali's father bought a new gold ornament. The shopkeeper claimed it was 22-carat gold. Anjali noticed that some metals are alloyed with gold to make ornaments.

(i) Why is pure gold (24-carat) not used to make ornaments? (1)

(ii) Name two metals usually alloyed with gold. (1)

(iii) What does '22-carat' indicate? (1)

(iv) Define an alloy. Give one common example apart from gold alloys. (1)

— End of Question Paper —